

Homework#5

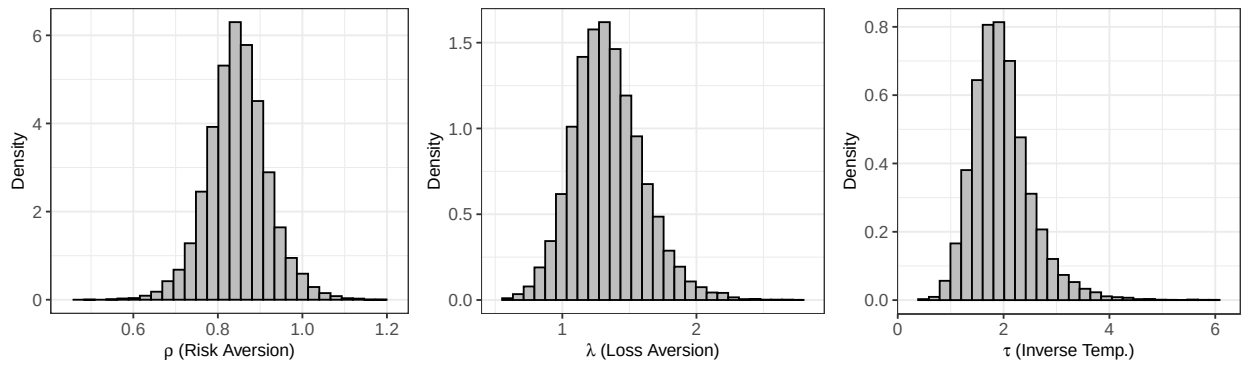
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Q1

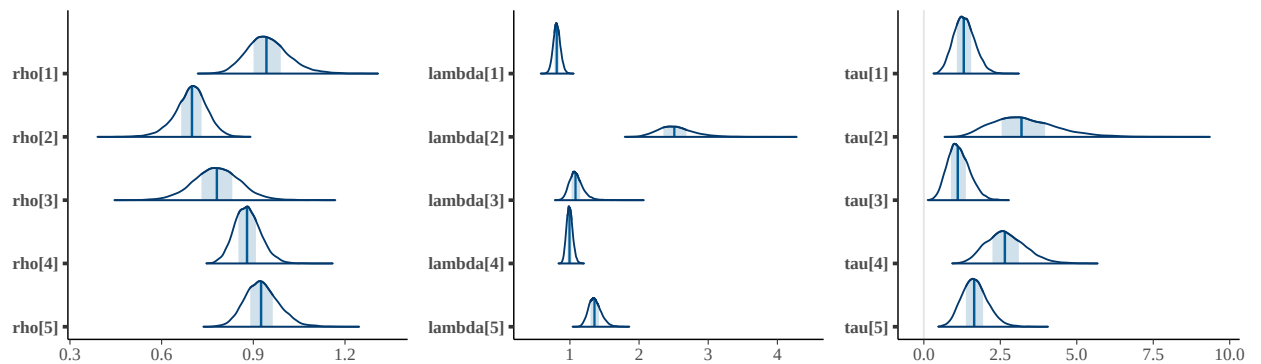
To run this code, run Q2/Q2_hBayesDM.R first.

(1) hBayesDM vs. MLE

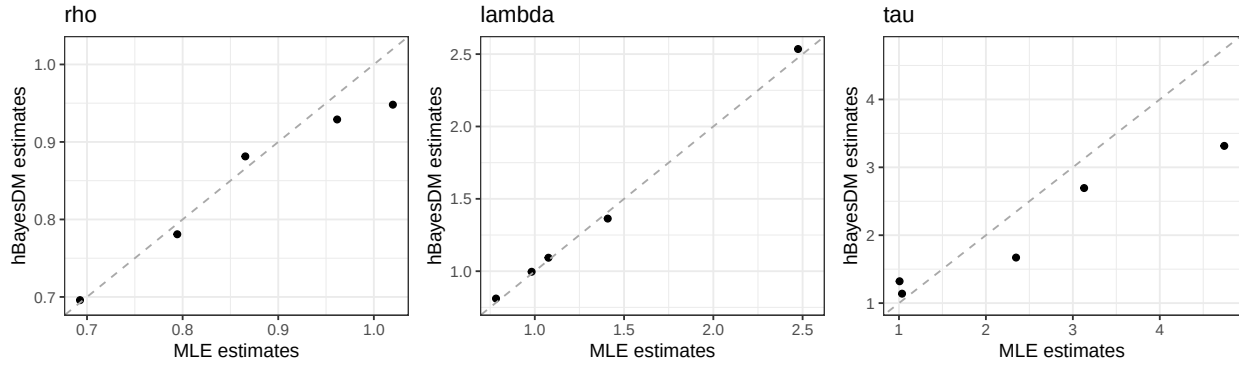
hBayesDM (group-level parameters)



hBayesDM (individual-level parameters)



(2) Visualization



(3) discussion

The hBayesDM uses hierarchical approaches, where individual-level estimates inform group-level estimates, and group-level estimates constrain individual-level estimates. This leads to “shrinkage” effect, where individual parameter estimates tend to be closer to the group-level estimates, compared to individual-level approaches of the stan results.

Q2

a) Bayes factor of two models

$$3.2445401 \times 10^{15}$$

b) evidence of model1

extreme evidence for model1 (ra_prospect model)

c) posterior odds

$$4.8668101 \times 10^{15}$$

d) posterior model probability of model 1

1

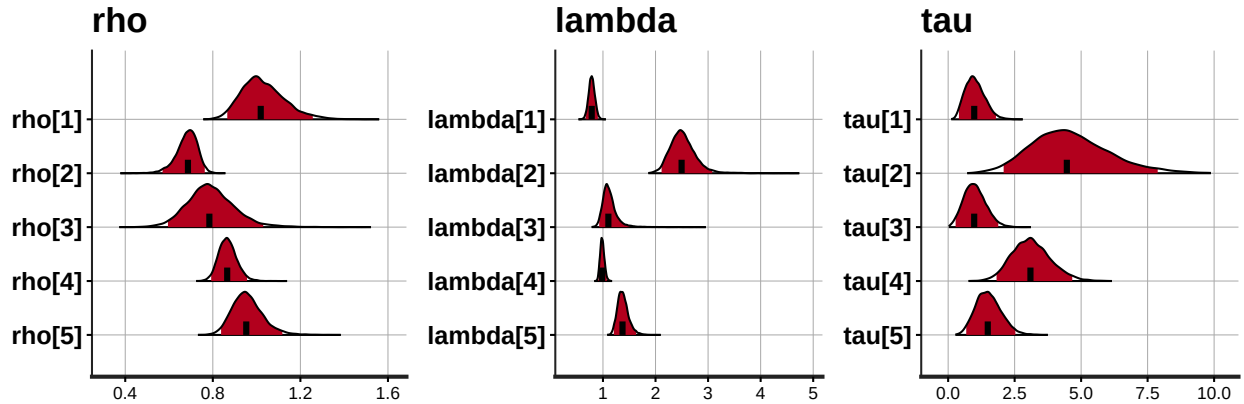
e) Bayes factors of seven memory retention models

Table 1: Bayes Factor (BF) Table

	POW1	POW2	EXP1	EXP2	EXPOW	HYP1	HYP2
POW1	1.00e+00	3.54e-01	2.03e-11	8.92e-04	1.25e-01	2.17e-02	4.20e-02
POW2	2.83e+00	1.00e+00	5.74e-11	2.52e-03	3.53e-01	6.13e-02	1.19e-01
EXP1	4.92e+10	1.74e+10	1.00e+00	4.39e+07	6.15e+09	1.07e+09	2.07e+09
EXP2	1.12e+03	3.96e+02	2.28e-08	1.00e+00	1.40e+02	2.43e+01	4.70e+01
EXPOW	8.00e+00	2.83e+00	1.62e-10	7.14e-03	1.00e+00	1.74e-01	3.36e-01
HYP1	4.61e+01	1.63e+01	9.36e-10	4.12e-02	5.76e+00	1.00e+00	1.93e+00
HYP2	2.38e+01	8.43e+00	4.84e-10	2.13e-02	2.98e+00	5.17e-01	1.00e+00

Q3

(1) Posterior distributions of all 5 subjects



(2) posterior means of 5 subjects

Table 2: Posterior means of each subject

subjID	rho	lambda	tau
2	1.031	0.786	1.011
3	0.681	2.521	4.590
4	0.792	1.129	1.006
6	0.868	0.987	3.129
7	0.959	1.383	1.521